PERMITPULSE

Marine Operations Work Permit - MVP Test Permit

UNOFFICIAL / SYNTHETIC TEST DOCUMENT

NOT A REGULATORY AUTHORIZATION	This document is for PermitPulse software testing and demonstration only. It does not constitute a marine licence, permit-to-work authorization, regulatory approval, legal compliance confirmation, or safe-to-work certification.

Permit ID	PP-MARINE-2026-2028-001
Operation	Marine dredging, disposal and vessel-support operations
Validity period	1 October 2026 00:00 UTC to 31 October 2028 23:59 UTC
Test position	54.025000 N, 5.925000 W (WGS84)

1. Defined operational position

Operations covered by this test permit shall be assessed at Latitude 54.025000 N, Longitude 5.925000 W, coordinate reference system WGS84.

Environmental evidence shall retain the actual provider/model point selected and the distance between that point and the permit position.

No alternative or midpoint location may be substituted without explicit review.

2. General evidence requirement

Before an operational period is assessed, PermitPulse shall use available reviewed evidence for the selected position and requested UTC period.

Missing, stale, unsupported, spatially inadequate, or temporally inadequate evidence shall result in UNKNOWN, not an assumed pass.

3. Significant wave height

Significant wave height (Hs) shall be assessed in metres.

STOP condition: Hs greater than 1.5 m.

Passing condition: Hs less than or equal to 1.5 m.

Source dataset, valid time, model point, and model-point distance shall be retained.

4. Mean wave period

Mean wave period shall be assessed in seconds.

STOP condition: mean wave period greater than 8.0 s.

Passing condition: mean wave period less than or equal to 8.0 s.

5. Wave direction

Mean wave direction shall be recorded in degrees using the provider direction convention.

Wave direction is informational only and shall not independently produce STOP.

6. Wind speed

10 m wind speed shall be assessed in metres per second.

STOP condition: 10 m wind speed greater than 12 m/s.

Passing condition: wind speed less than or equal to 12 m/s.

7. Wind gust

Wind gust shall be assessed independently of sustained wind speed.

STOP condition: wind gust greater than 18 m/s.

Wind gust shall not be estimated from sustained wind using an arbitrary multiplier.

8. Wind direction

10 m wind direction shall be recorded in degrees and retain the provider direction convention.

Wind direction is informational only and shall not independently trigger STOP.

9. Current speed

Surface-current speed shall be assessed in metres per second.

STOP condition: current speed greater than 1.0 m/s.

Where eastward and northward components are available, current speed may be calculated deterministically from those components.

10. Eastward current component (U)

Eastward current component U shall be retained separately.

STOP condition: absolute eastward component greater than 0.60 m/s.

The signed source value shall also be preserved.

11. Northward current component (V)

Northward current component V shall be retained separately.

STOP condition: absolute northward component greater than 0.60 m/s.

The signed source value shall also be preserved.

12. Current direction

Current direction may be derived deterministically from retained U/V components.

Direction shall be expressed as TOWARDS, clockwise from true north.

Current direction is informational and shall not independently trigger STOP.

13. Visibility

Visibility shall be assessed only where supplied explicitly by an accepted evidence source.

STOP condition: visibility below 1,000 m.

Passing condition: visibility greater than or equal to 1,000 m.

Visibility shall not be inferred from humidity, rainfall, cloud, or other proxy variables.

14. Turbidity

Turbidity shall be assessed in FNU using compatible Copernicus Marine turbidity evidence, including observation or modelled products, whose native or verified normalized unit is FNU.

STOP condition: turbidity greater than 25 FNU.

Passing condition: turbidity less than or equal to 25 FNU.

Evidence reported only in NTU, or in another incompatible unit, shall not be treated as equivalent to FNU. Such evidence shall result in UNKNOWN unless an explicitly reviewed compatible conversion or threshold exists.

The evidence record shall retain source product/dataset, variable, evidence mode (observation or model), native unit, normalized unit, valid time, selected evidence point, and point-distance information.

15. Dissolved oxygen

Dissolved oxygen shall be assessed at the **sea-surface model level (0 m below sea surface)** using compatible Copernicus Marine modelled dissolved-oxygen evidence. Unit: mg/L.

STOP condition: dissolved oxygen below 6.0 mg/L.

Passing condition: dissolved oxygen greater than or equal to 6.0 mg/L.

The evidence record shall retain requested depth, actual provider/model depth, source unit, normalized unit, and valid time.

If available evidence does not include a compatible 0 m model level, the result shall be UNKNOWN. Modelled dissolved oxygen shall not be represented as equivalent to an in-situ measurement.

16. Continuous work-window requirement

An operational period may be classified as a candidate WORK interval only where all applicable decision-relevant conditions are sufficiently evidenced and pass for the entire requested uninterrupted work duration.

Example: if a 6-hour operation is requested but acceptable conditions persist for only 5 hours, the period shall not qualify as a 6-hour WORK window.

17. Decision states

STOP - one or more applicable reviewed constraints are demonstrably triggered.

UNKNOWN - no STOP has been established, but required evidence, review, applicability, coverage, or compatibility remains unresolved.

WORK - every applicable decision-relevant constraint has sufficient evidence and passes for the complete evaluated interval.

Decision precedence: STOP > UNKNOWN > WORK.

18. Evidence limitations

PermitPulse shall not invent missing measurements.

PermitPulse shall not infer visibility from weather proxies.

PermitPulse shall not infer wind gust from sustained wind.

PermitPulse shall not automatically convert NTU to FNU.

PermitPulse shall not substitute sea-surface height for local tide-gauge evidence.

PermitPulse shall not substitute an incorrect dissolved-oxygen depth.

PermitPulse shall not treat provider availability as evidence.

PermitPulse shall not treat a forecast as regulatory authorization.

19. Permit validity

This synthetic permit is valid only from **1 October 2026 00:00 UTC** through **31 October 2028 23:59 UTC**.

No evaluation outside this interval shall produce a WORK result under this test permit.

20. Operational-assessment limitation

A PermitPulse WORK result means only that the applicable reviewed constraints represented in the software passed against the available accepted evidence for the evaluated interval.

It does not constitute regulatory approval, legal compliance confirmation, vessel-master approval, permit-to-work authorization, or certification that operations are safe to perform.

Quick-reference threshold table

Parameter	Operational condition
Significant wave height	STOP if > 1.5 m
Mean wave period	STOP if > 8.0 s
10 m wind speed	STOP if > 12 m/s
Wind gust	STOP if > 18 m/s
Current speed	STOP if > 1.0 m/s
Eastward current |U|	STOP if > 0.60 m/s
Northward current |V|	STOP if > 0.60 m/s
Visibility	STOP if < 1,000 m
Turbidity (Copernicus-compatible FNU)	STOP if > 25 FNU
Dissolved oxygen at 0 m model level	STOP if < 6.0 mg/L
Wave / wind / current direction	Informational only

Modification note: Conditions 14 and 15 were revised for PermitPulse provider-compatibility testing. The permit validity start date is aligned to 1 October 2026 to match the current synthetic acceptance scenario.

End of synthetic PermitPulse MVP test permit.